

Recovery Lab

TVS Credit EPIC 8 | Executive summary | 6 September 2026

A local proof of concept that estimates the sale value of a repossessed vehicle, displays a calibrated uncertainty range, and compares lending offers under explicit collateral and affordability assumptions. A reviewer retains the decision.

Measured result on the final operational holdout

Measure	Final result
Operational test cases	1,244
Model / baseline mean absolute error	INR 7,052 / INR 8,343
Reduction in mean absolute error	15.5%
Calibrated interval coverage	80.5% (1,001 of 1,244)
Mean interval width	INR 23,146

These are internal development holdout results. The model and calibration remained frozen. Operational seizures span 1 May to 9 July 2026, with observed sales from 20 June to 31 July. The sold-only sample and incomplete later outcomes limit generalization.

Data, method and leakage controls

The original Analytics Case Study Dataset.xlsx remains unchanged. The audit retained all 15,000 records and 41 source fields, with eight derived fields. Training used 9,767 cases, model selection 576 and calibration 583, with sequential label cutoffs before January, March and May 2026. Nine vehicle and location inputs exclude sale outcomes, yard duration and unresolved condition codes. Agreement duration is not manufacture age.

CatBoost MultiQuantile estimates lower, median and upper values. A frozen nonnegative conformal correction expands each interval side by INR 1,859. SHAP contributions describe model associations. The 80% target is pooled and does not establish coverage for each segment or future period.

The EV limitation changes the proposed rollout

Only 22 of 36 EV cases fall inside the range: 61.1% coverage. EV mean error is INR 13,509, just 0.4% better than its baseline, with average underprediction of INR 7,260. The EV sample is small. More EV outcomes and independent validation are required before operational reliance. Petrol coverage is 81.0% on 1,208 cases.

Decision workflow and pilot gates

Business demonstration

The policy simulator checks 35 loan offers against a 40% payment-to-income cap and a 20% downside shortfall cap. It ranks feasible offers by highest principal, then lowest contractual interest, then shorter term. Income and existing obligations require verification. The demonstration policy does not represent an approved TVS Credit policy.

For fictional reference A (INR 100,000 cost, INR 70,000 downside anchor, INR 85,000 median anchor, INR 30,000 monthly income and INR 5,000 existing payments), base assumptions yield INR 80,000 for 18 months. Adverse assumptions yield INR 70,000 for 12 months. Severe assumptions yield no feasible offer. These hand-authored anchors remain separate from actual model predictions.

A separate fixed three-loan fictional book has INR 265,000 starting exposure. Under month-zero default for all loans, shortfall rises from INR 62,000 (23.4%) in base conditions to INR 121,880 (46.0%) in severe conditions. Asset shortfalls are summed without offsetting surplus collateral. This is conditional severity, not default probability or expected loss. Future depreciation factors are assumptions, not validated forecasts.

Local delivery

One Python service serves the compiled dashboard at <http://127.0.0.1:8078/>. Strict input validation, bundle hashes, readiness checks, rate limits and a persistent SQLite audit journal support the demonstration. If the primary model fails, the service returns a baseline for review, removes the calibrated interval and disables model-based lending.

The recorded Phase 8 benchmark had a serial p95 of 84 ms over 24 local requests, including prediction, explanation and audit handling. This small loopback test does not establish production capacity. The package includes source code, frozen models, an executed notebook, data dictionary, model card and a five-minute demonstration script.

Conditions for a production pilot

Collect prospective outcomes including unsold vehicles, full recovery costs and verified feature timestamps. Confirm CIBIL sentinel meanings, salary frequency, condition definitions and borrower obligations. Validate EV performance separately. Add organizational identity and access controls, approved policy, backup/restore procedures, outcome monitoring and a governed recalibration process. Use a fresh forward test after model changes.

The secondary retrospective analysis contains 1,835 cases, including all 1,244 operational cases, so it is not an independent replication. The largest observed error has an actual sale of INR 15,000 versus a median estimate of INR 61,100. Its cause remains unknown. No causal profit increase or hackathon outcome is claimed.

Sources and implementation credit

Evidence: supplied case study and workbook; Phase 1 audit; Phase 2 split manifests; Phase 8 operations and latency reports; Phase 9 `final_metrics.json`, `operational_segments.json` and `Model_Card.md`. The accompanying notebook reproduces frozen inference and measured metrics.

Method references: CatBoost regression objectives, [official documentation](#); Romano et al. (2019), [Conformalized Quantile Regression](#). CatBoost, FastAPI and React provide libraries. The POC integrates time controls, policy calculations, scenario accounting and evidence reporting. Adaptive online calibration is not implemented.